İshak Alkuş

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About Me

Senior Embedded Systems Engineer with over 10 years of expertise in delivering full-cycle IoT solutions, from hardware architecture to cloud integration. Proven track record of deploying 50+ commercial products in industrial automation, environmental monitoring, and smart metering. Specialist in low-power wireless design (LoRaWAN, BLE, Wi-Fi, NB-IoT and LTE-M), high-precision analog systems, and battery management. Expert in developing robust firmware, custom hardware abstraction layers, and companion desktop/web applications for end-to-end system control.

Core Technical Skills

Programming Languages: C, C++, C#, Python, JavaScript (partial)

Microcontrollers: STM32 (HAL/LL), ESP32 (ESP-IDF), PIC, nRF Series

Wireless Protocols: LoRaWAN, WiFi, BLE, ESP-NOW, NB-IoT & LTE-M

Communication Protocols: ModBus RTU/TCP, RS485, UART, I2C, SPI, CAN, HTTP, MQTT, WebSocket, CoAP, LwM2M

Development Environments: VS Code, STM32CubeIDE, Keil µVision, Segger Embedded Studio

Operating Systems: FreeRTOS, Zephyr, Linux (Embedded, Yotco), Bare-metal programming

PCB Design: Eagle (Primary), KiCad, Altium Designer

DevOps & Deployment: Docker, systemd, Raspberry Pi, Git

Test Equipment: JTAG/SWD Debuggers, Oscilloscopes, Logic Analyzers, Power Analyzers, SDRs

Professional Experience

Senior Embedded Systems Engineer | CRATUS - Remote

2022 - Present

Dual Wireless CAN-BUS Bridge

- Designed and implemented a wireless connection bridge for CAN devices on the field where cabling is not possible. The device uses a proprietary communication protocol to reach 500 meters even in metal dense environments. The device also contains a Wi-Fi controller to be able to transfer the CAN messages to the cloud.

LOMA AI Hardware Abstraction Layer

- Architected a production-grade Hardware Abstraction Layer bridging Modbus control boards with cloud microgrids. Simplified thousands of register mappings into a service-oriented architecture with bidirectional WebSocket integration, significantly reducing integration time for non-embedded developers.

High-Speed UART to WebSocket Bridge

- Designed and implemented UART to WebSocket bridge with extremely high data rates using ESP32 and STM32.

Modbus to MQTT Bridge with Multiple Connectivity Options

- Built a complete Modbus to MQTT bridge, fully customizable and supporting various connectivity options such as Ethernet, Wi-Fi, NB-IoT and LTE-M. The device supports over the air configuration for register mapping and port configurations. Provisioning and credential setting can be done in the field by user or pre-configured.

Smart BESS Calculator Web Application

- Built a full-stack web application (React/Node.js) for Battery Energy Storage System planning, calculating ROI and energy consumption based on real-time climate and location APIs.

IoT Device Architecture & Firmware Development

- Developed 10+ different WiFi/BLE connected devices for smart home and industrial applications
- Created various custom firmware libraries for sensors, communication modules, and peripheral integration
- Designed modular "Lego-style" IoT platform allowing mix-and-match connectivity (BLE, LoRaWAN, NB-IoT)
- Led cross-functional teams in complex embedded projects

Manufacturing & Testing Infrastructure

- Developed manufacturing and test software using C#, Python, and JavaScript
- Implemented Over-The-Air (OTA) update mechanisms for remote firmware deployment for MCU families like STM32, NRF52 and ESP32.
- Created automated testing frameworks ensuring product quality and reliability

High Voltage DC Systems (HVDC)

- Engineered BMS solutions for applications ranging from 24V small-scale to 800V high-voltage systems as planned the system components, generated connection diagrams and installation manuals, planned and implemented communication scheme between different system components and added a custom data acquisition platform.
- Implemented modular, scalable management architectures for EV and energy storage applications

Soil Thermal Conductance Measurement System

- Developed a precision measurement device utilizing a high-grade analog front-end and ESP32 (ESP-IDF, FreeRTOS). Implemented robust FUOTA (Firmware Update Over The Air) via both BLE and USB. The device is currently active in the field, delivering high-accuracy data.

Battery Gauge ASIC Simulation Platform

- Engineered a complex development bed for battery gauge ICs using multiple synchronized STM32 MCUs. Integrated high-precision AFEs and complex data acquisition algorithms. Developed a custom .NET/C# desktop application for real-time logging, control, and visualization.

Wildfire Detection Sensor

- Designed a LoRaWAN-connected forest safety node featuring GNSS, CO, and CO2 monitoring. Implemented adaptive learning algorithms to filter false positives, achieving a battery life of 10+ years without maintenance.

Embedded Systems Engineer | SKYSENS

2018 - 2021

LoRaWAN Network Solutions

- Developed 20+ different LoRaWAN device types for environmental and industrial monitoring
- Successfully deployed across 50+ business sites including BRİSA, Istanbul Airport, Kale Factories, and Vestel
- Achieved 20+ km communication range in rural environments
- Optimized power management to deliver 10+ years battery life using single AA-size LiSOCl2 batteries

Industrial Integration & Innovation

- Designed OTA programmable ModBus master device integration for industrial control systems working with LoRaWAN class c.
- Developed specialized solutions for magnetic car park detection and wildfire monitoring
- Built manufacturing test suites and update systems using Python and C# for production scalability

Research Engineer | TÜBİTAK & İzmir Katip Çelebi University

2016 - 2018

- Led research project: "Robust Adaptive Controller Design and Applications with Guaranteed Online Stability"
- Designed and simulated adaptive control systems using MATLAB & Simulink
- Tested algorithms on drones, inverted pendulums, chaotic mixers, and robotic systems
- Published 2 peer-reviewed conference papers on adaptive control applications

Leadership & Client Management

- Successfully managed 30+ embedded systems projects from concept to deployment
- Guided and trained multiple junior engineers and fresh graduates
- Direct technical consultation and support for complex IoT and Industrial implementations
- Successfully performed extensive client support and technical consultation for many clients throughout project lifecycles.
- Led interdisciplinary teams in complex technical projects

Education

M. Eng. Electronics Engineering | Ege University | 2018-2020 (not completed due to COVID-19)

B.S. Electrical & Electronics Engineering | İzmir Katip Çelebi University | 2012-2016

Exchange Program - Telecommunications Engineering | Poznan University of Technology | 2014